

Ranking Calgary's water mains for replacement

Gaurav Kumar · Track C — Classical ML · Own domain, framed nearest S1 (detect supply disruptions early) · Repo: github.com/gauravk09/calgary-water-mains

1. Problem & domain

Calgary runs ~5,400 km of water mains. Every break is repaired as an emergency, but the city can only **replace** ~8–15 km a year (0.15–0.3% of the network). Each January the asset team must choose which pipes to replace so the most future breaks are avoided. A break is a disruption of the water supply network; the shortlist is preventive maintenance of that supply chain. The data is real and open (City of Calgary): 37,538 breaks since 1956 with failure mode and location, and 60,740 pipe segments with material, diameter, install year and length. Chosen after screening ~20 public datasets for one whose value can be measured without invented costs: **breaks avoided per km replaced**.

Why Track C: structured attributes and event history, a rare count target, and explanations an engineer can read — classical models with SHAP fit; LLM tracks add cost without signal.

2. Approach & algorithm decisions

Decision	Choice	Alternative considered → why rejected
Unit & label	Pipe × 1 January; label = breaks on that pipe during the year (all failure modes)	Yes/no → a pipe breaking 3× avoids 3 breaks; counts scale cleanly with length
Linking	Breaks carry no pipe ID → nearest pipe (99% within 1.3 m); keep breaks on pipes still in the ground	All breaks → 59% are on pipes since replaced, absent from the inventory
Model	LightGBM, Poisson objective, log(length) offset , regularised (min leaf 2,000, L2 10, tuned on validation) → breaks per km-year	Poisson GLM → over-predicts counts 23–33% (misses material × history interactions). XGBoost → same family, tied. Isolation Forest → finds <i>unusual</i> pipes, not risky ones (≈0 recall on never-broken pipes). Neural net → boosted trees usually match it on tabular data, with clearer explanations
Features	Material, zone, diameter, age, length; own history (count, per km, recency); breaks on other pipes within 50/150/300 m , incl. since-replaced pipes — all before 1 January	Past failure types → no gain; break coding changed around 2000
Split	Sliding window: for each year train on all earlier years. Validation 2016–20 (every choice), test 2021–25 (once)	5-year snapshots → tied but stale. 4-year window → worse (6.9 vs 7.3% R@1%); history beats recency
Metric	Recall at 0.25% of network length per year (one year's budget), pooled over 5 years; lift = recall ÷ budget	Accuracy → 99.6% of pipe-years have no break. AUC → averages over budgets no city funds
Calibration	Raw predicted counts	Rescale by last year's actual/predicted → worse on validation (0.96)

Assumptions and how they were checked: breaks follow a rate proportional to length (validation predicted/actual = 1.04); the nearest pipe is the broken pipe (distance distribution); replacement prevents all failure modes on the segment and cost scales with length (stated, not testable here). **Imbalance:** 99.6% zeros are handled by modelling counts with an exposure offset rather than resampling, which

keeps predicted counts calibrated. **Lessons:** seed-averaged comparisons (one seed flipped a ranking); a train-vs-held-out check showed memorising (lift $27\times$ vs $8.8\times$) → regularisation cut it to $15.5\times$ vs $8.8\times$.

3. Results & error analysis (test 2021–2025, 855 breaks)

Test years 2021–2025 (855 breaks): breaks caught by each replacement list

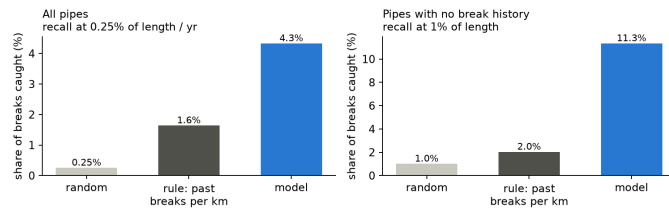


Fig 1. Breaks caught: random vs rule vs model. Lift at the real budget: rule $6.6\times$, model $17.3\times$.

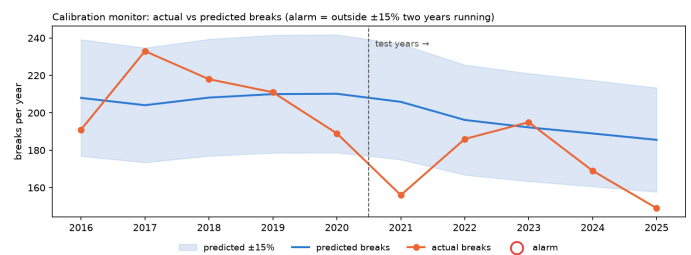


Fig 2. Yearly calibration monitor (Evidently test, $\pm 15\%$): 2021 and 2025 fail; alarm needs two in a row.

Business impact: at Calgary's real budget (0.25% of length a year) the model catches 4.3% of breaks vs 1.6% for the rule — **~37 vs 14 breaks avoided over five years**, four to five fewer emergency digs a year for the same capital spend. R@1% is 10.4% overall and 11.3% on never-broken pipes (rule 2.0%).

Validation lift was $9.0\times$; one year's budget catches 3–14 breaks, so the honest range is **9–17 $\times$ random**. SHAP (Fig 3) shows material dominates, then recent breaks nearby, then the pipe's own record; among cast iron, risk *falls* with age past ~60 years (the oldest survivors are the sturdier remainder), so "oldest first" is the wrong rule.

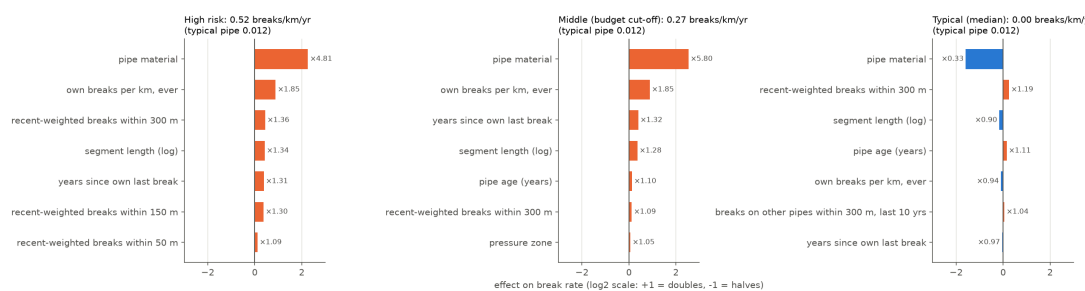


Fig 3. Why three pipes rank where they do (SHAP as multipliers on a typical pipe's break rate): cast iron, its own break record and recent breaks nearby lift the top pipes; PVC pulls the typical pipe down.

Where it fails: (1) 85% of pipes that broke in 2021–2025 broke only once, often on pipes indistinguishable from their neighbours — much of the gap to perfect foresight is chance, not missing modelling; (2) the top of the list is dominated by repeat offenders, where the simple rule is already as good; the model's gain comes from pipes with no history; (3) **calibration on test: predicted/actual = 1.14** — break rates kept falling faster than the model expected.

4. Production & limitations

Production: retrain every 1 January on all history, publish the ranked shortlist with per-pipe SHAP reasons, and let the engineer add local knowledge before approving. Monitoring covers two concrete failure modes: **(a) falling break rates** → **over-prediction**, caught by a yearly Evidently test (actual within $\pm 15\%$ of predicted; alarm after two failing years) — replayed over 2016–2025 it fails in 2021 and 2025; a second consecutive failure would trigger a retrain or rescale; **(b) shifting inputs** (ageing network, recording changes) via an Evidently feature-drift report — it flagged pipe age and nearby breaks but *not* the label, because mostly-zero labels hide rate changes, which is why (a) is needed. A pytest test proves features cannot see future breaks.

Limitation to fix before deployment: the pipe inventory is a 2026 snapshot, so every historical training year only contains pipes that survived to 2026 — the worst pipes, already replaced, are missing, which likely flatters the test score. Historical inventories (or the replacement log) are needed; repair-cost data would also let breaks be weighted by severity and turned into dollars.